

NAME - PRERANA MUDIAR
ROLL NO - CSB24021

```
#include <stdio.h>
#include <stdlib.h>

void constantSpace(int n) {
    int a = 10; // only one variable
    printf("Constant Space (n=%d): %lu bytes\n", n, sizeof(a));
}

void linearSpace(int n) {
    int *arr = (int *)malloc(n * sizeof(int)); // 1D array
    printf("Linear Space (n=%d): %lu bytes\n", n, n * sizeof(int));
    free(arr);
}

void quadraticSpace(int n) {
    int **arr = (int **)malloc(n * sizeof(int *));
    for(int i = 0; i < n; i++)
        arr[i] = (int *)malloc(n * sizeof(int)); // 2D array

    printf("Quadratic Space (n=%d): %lu bytes\n", n, n * n * sizeof(int));

    for(int i = 0; i < n; i++)
        free(arr[i]);
    free(arr);
}

int main() {
    int n;

    printf("Enter input size: ");
    scanf("%d", &n);

    constantSpace(n);
    Linear space(n);
    Quadratic space(n);

    Return 0;
}
```

Output

```
Enter input size: 1000
O(1) space for n=1000 -> 12 bytes
O(n) space for n=1000 -> 4000 bytes
O(n^2) space for n=1000 -> 4000000 bytes
```